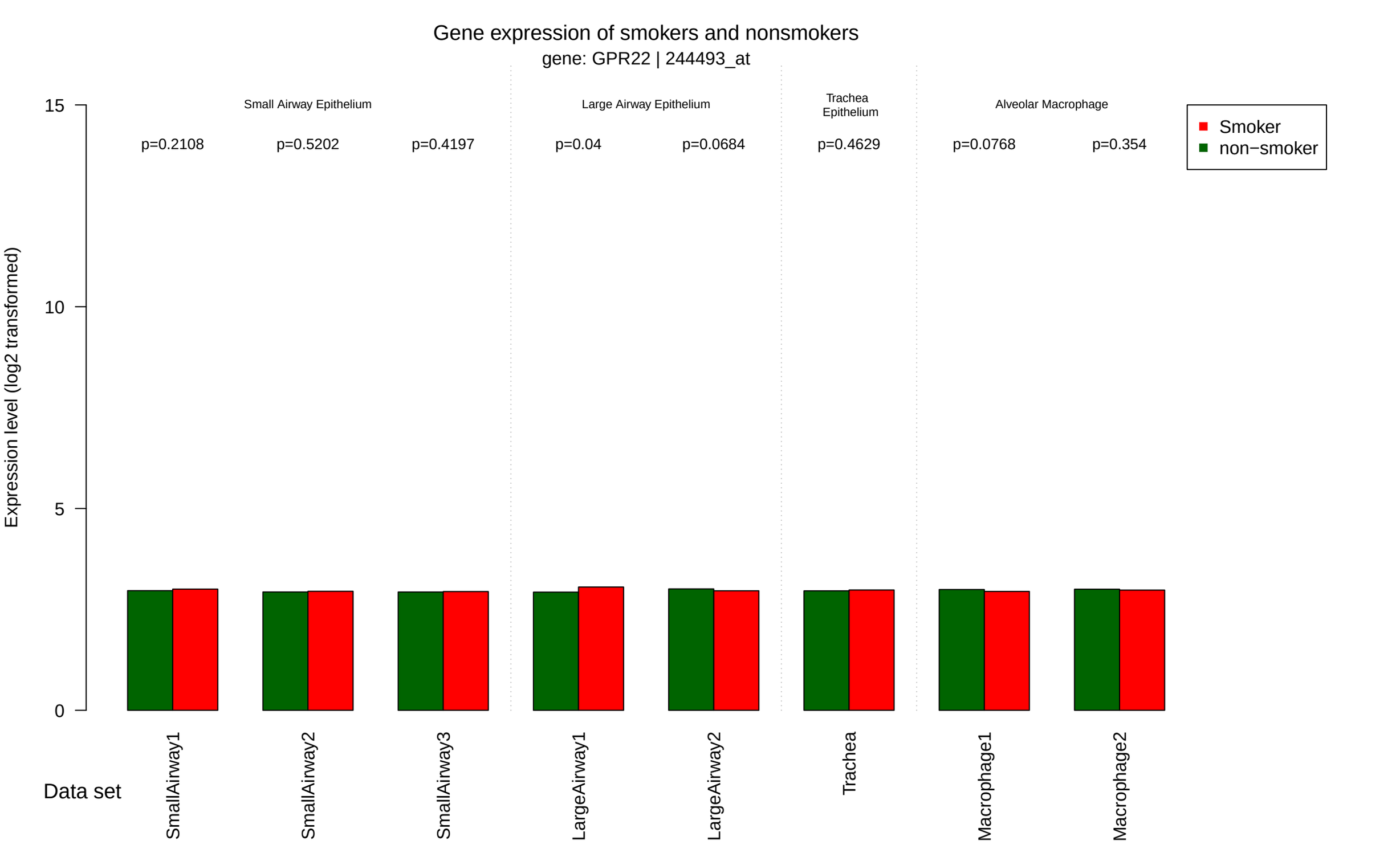
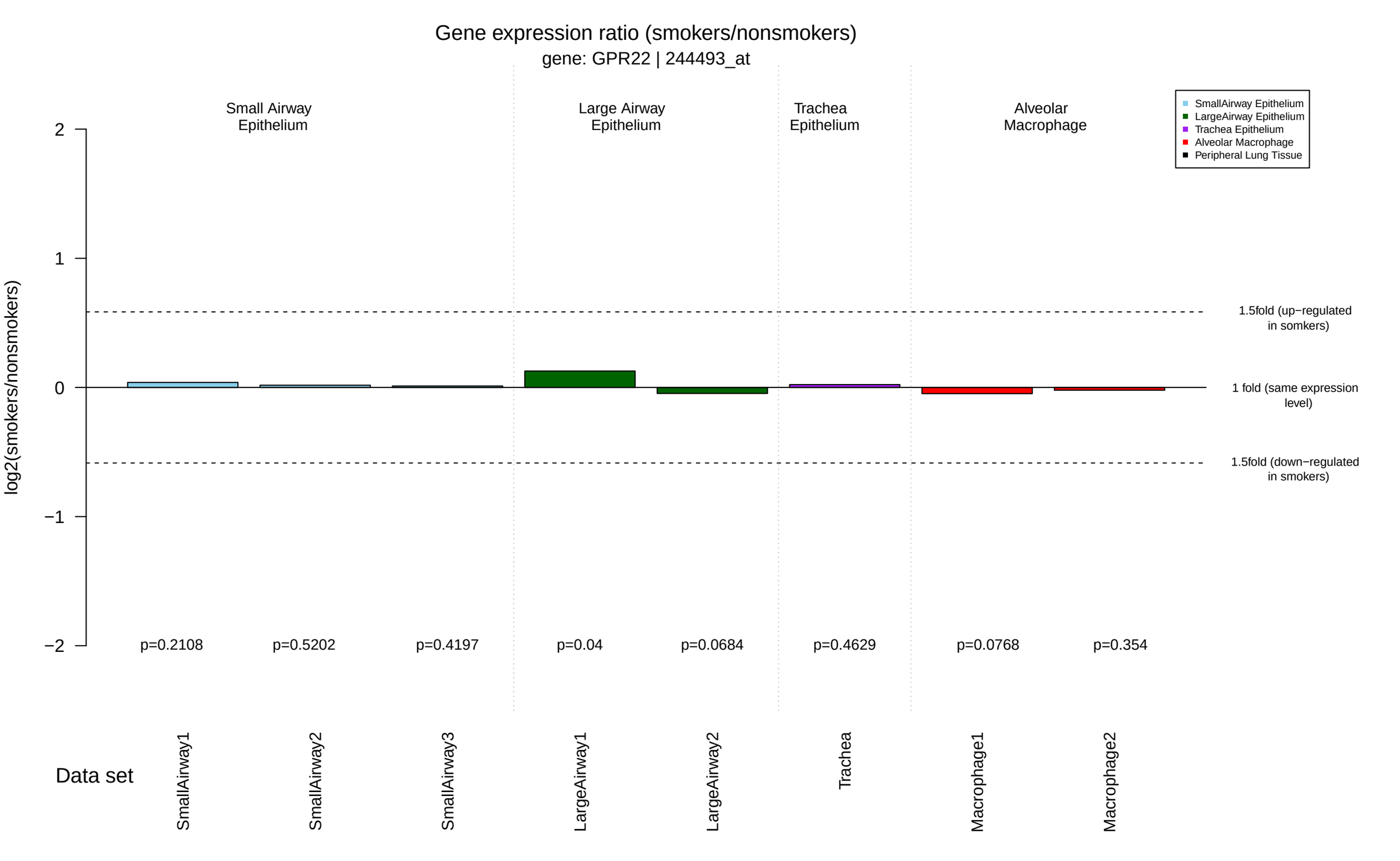
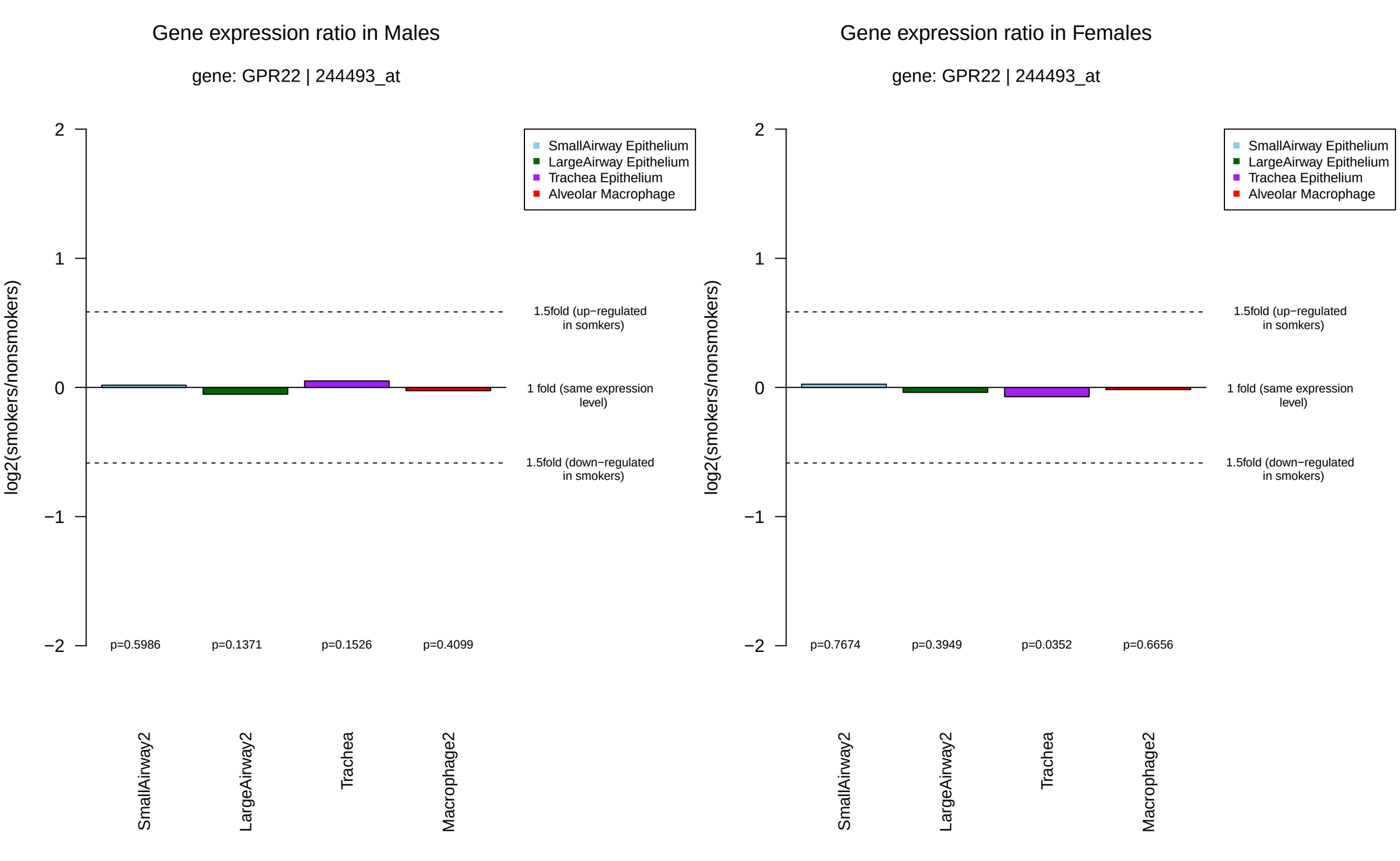
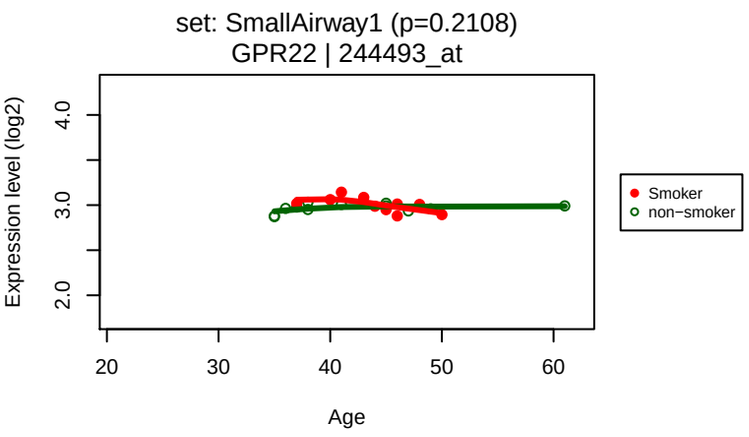
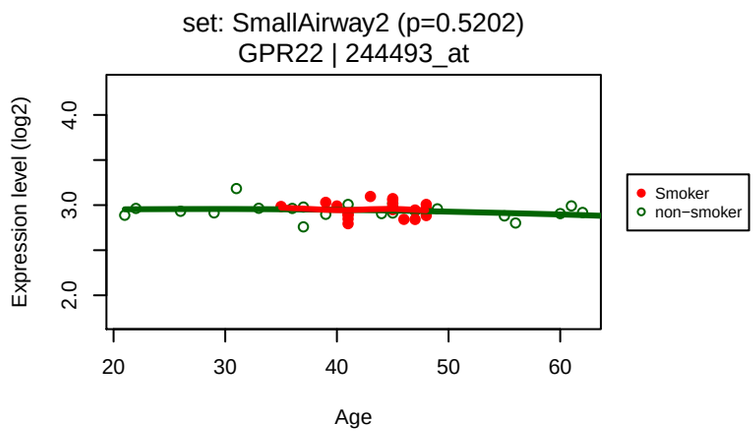
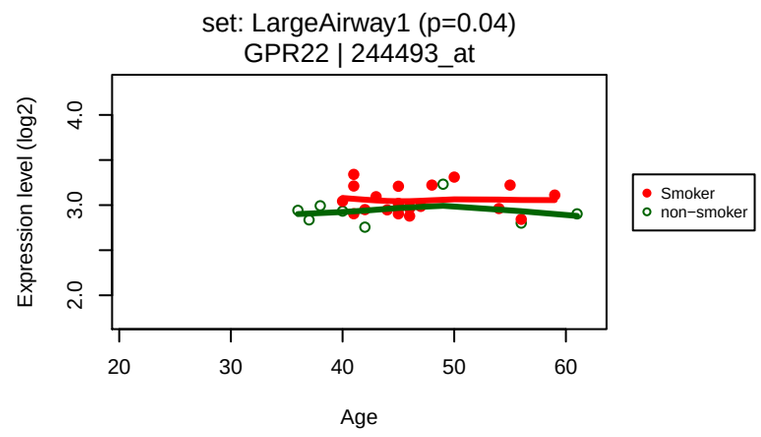

Gene expression of smokers and nonsmokers
gene: GPR22 | 244493_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2108
p=0.5202
p=0.4197
p=0.04
p=0.0684
p=0.4629
p=0.0768
p=0.354
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

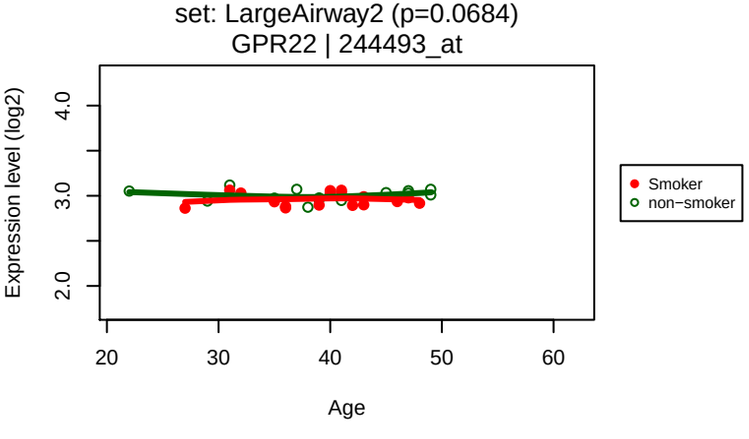

Gene expression ratio (smokers/nonsmokers)
gene: GPR22 | 244493_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2108
p=0.5202
p=0.4197
p=0.04
p=0.0684
p=0.4629
p=0.0768
p=0.354
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

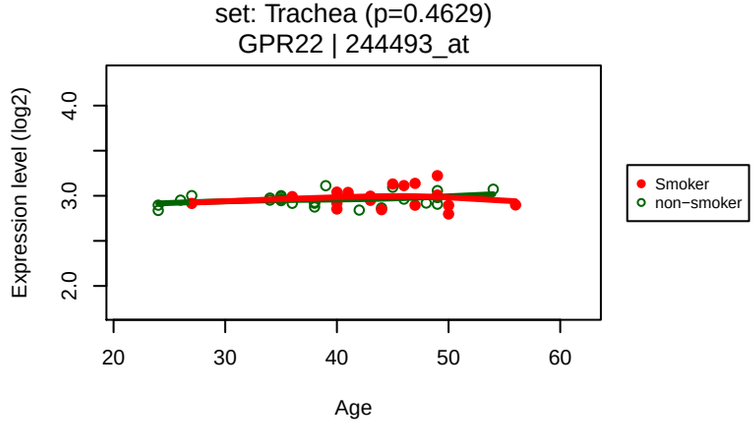

Gene expression ratio in Males
gene: GPR22 | 244493_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5986
p=0.1371
p=0.1526
p=0.4099
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR22 | 244493_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7674
p=0.3949
p=0.0352
p=0.6656
SmallAirway2
LargeAirway2
Trachea
Macrophage2

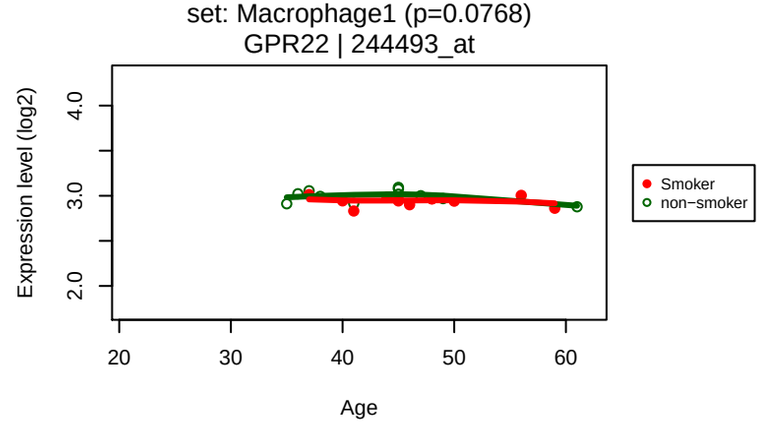

set: SmallAirway1 (p=0.2108)
GPR22 | 244493_at
Expression level (log2)
Age
Smoker
non−smoker

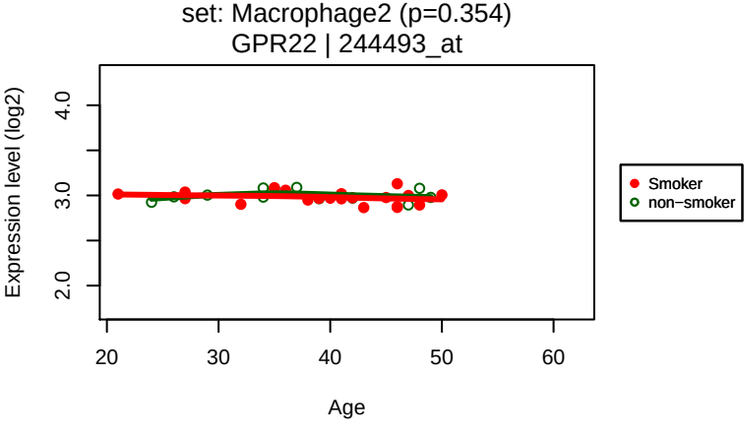

set: SmallAirway2 (p=0.5202)
GPR22 | 244493_at
Expression level (log2)
Age
Smoker
non−smoker

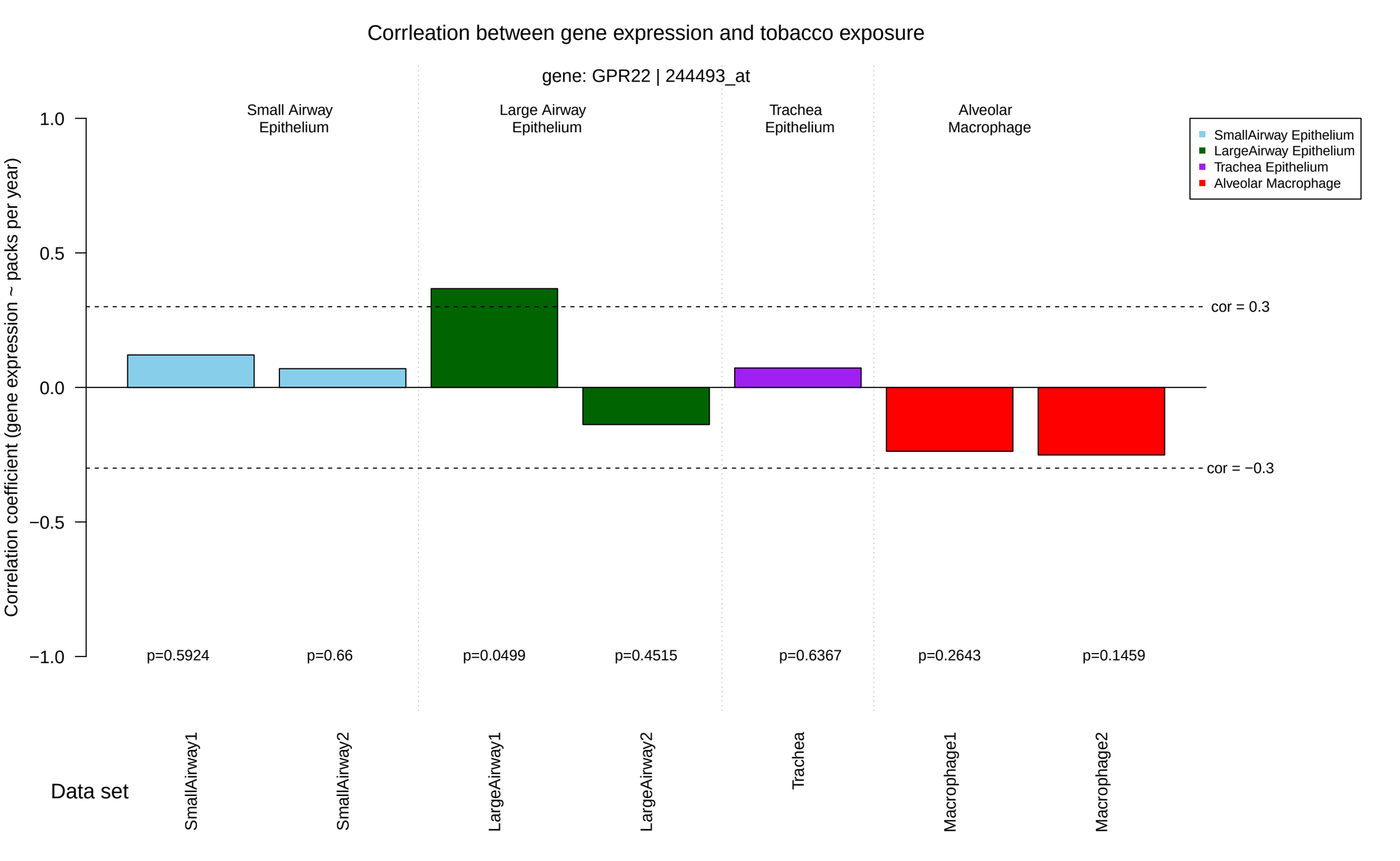

set: LargeAirway1 (p=0.04)
GPR22 | 244493_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0684)
GPR22 | 244493_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4629)
GPR22 | 244493_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0768)
GPR22 | 244493_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.354)
GPR22 | 244493_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR22 | 244493_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5924
p=0.66
p=0.0499
p=0.4515
p=0.6367
p=0.2643
p=0.1459
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2